BITS / HEX

- k bits → 2^k values; N items → ceil($\log_2 N$) bits
- 1 hex digit = 4 bits; n hex digits = 4n bits = n/2 bytes
- A=1010 B=1011 C=1100 D=1101 E=1110 F=1111
- hex addr n digits → 2^(4n) bytes addressable

0x3ACF → 16 bits, 2 bytes, 64KB addressable
0x7F3A = 0111 1111 0011 1010

Dec→hex: divide by 16, read remainders bottom-up
185÷16=11(B) r9 → **0xB9** | 255÷16=15(F) r15(F) → **0xFF**
To verify: 0xB9 = 11×16+9 = 176+9 = 185 ✓

TWO'S COMPLEMENT

- k-bit signed: −2^(k−1) to 2^(k−1)−1 | 8-bit: −128 to 127
- MSB=1: signed = unsigned_val − 2^k
- Negate: flip all bits, add 1
- **Overflow:** two positives → negative result (or vice versa)

95+60=155 > 127 → overflow. 10011011_2 → 155−256 = **−101**
120+15=135 → 10000111_2 → 135−256 = **−121**
11111110_2 unsigned=254 → signed=254−256=**−2**

CUSTOM 6-BIT FLOAT

Layout: [s | e_1e_0 | $f_2f_1f_0$] (sign | 2-bit exp | 3-bit frac)

```
value = (-1)^s × 2^(exp-1) × (1 + f₂×0.5 +
f₁×0.25 + f₀×0.125)
```

010110: s=0, exp=10_2=2, scale=2^1=2, frac=1+0.5+0.25=1.75 → **3.5**

FP bias (general): bias = 2^(e−1)−1 where e = # exp bits
FP32: bias=127 | FP16 (5 exp): bias=15 | 4-bit exp: bias=7
BF16: 8 exp bits → bias=127, wider range than FP16

IEEE 754 FP32: DECIMAL → 32-BIT

Layout: [s (1b) | exp (8b, bias=127) | mantissa (23b)]

```
1. Sign: s=0 (+), s=1 (-)
2. Convert |x| to binary (int part + fraction
part)
3. Normalize: shift binary point → 1.xxx × 2^n
4. Stored exp = n + 127  (write as 8-bit
unsigned)
5. Mantissa = first 23 bits of xxx (zero-pad
right)
```

12.375: s=0 | 12=1100_2, 0.375=0.011_2 → 1100.011_2
→ 1.100011×2^3 | exp=3+127=130=10000010_2
→ 0 10000010 10001100000000000000000
Decode: 0 10000010 10001100... → exp=130−127=3
→ 1.100011×2^3 = 1100.011_2 = **12.375**

Frac→binary: 0.375×2=0.75(0), 0.75×2=1.5(1), 0.5×2=1.0(1) → .011

INSTRUCTION ALIGNMENT

- b-byte instructions: addresses are multiples of b
- # instructions = mem_size / b
- fraction valid byte addresses = 1/b

8-byte instr, PC=0x4000: next → 0x4008, 0x4010, 0x4018...
2KB memory: 2048/8 = **256 instructions** | valid: 1/8
4-byte ARM, 1KB: 1024/4=**256 instr** | valid: 1/4

POWERS OF 2 — QUICK LOOKUP

```
2^0=1        2^1=2        2^2=4        2^3=8
2^4=16       2^5=32       2^6=64       2^7=128
2^8=256      2^9=512      2^10=1,024   2^11=2,048
2^12=4,096   2^13=8,192   2^14=16,384 2^15=32,768
2^16=65,536 2^17=131,072 2^18=262,144
2^19=524,288
2^20=1,048,576 (≈1M)   2^24≈16M    2^30≈1B
(1GB)

Key: N bits → 2^N values → need
ceil(log₂(N_items)) bits
  70,000 items: 2^16=65536<70000<2^17=131072 →
17 bits
  4KB page = 2^12 → offset=12 bits
  1MB = 2^20 | 1GB = 2^30 | 4GB = 2^32
```

CPI / CPU PERFORMANCE

```
CPI_base = Σ(frac_i × cycles_i)
CPI_stall = CPI_base + mem_frac × stall_per_op
total_cycles = IC × CPI
time = total_cycles / clock_rate
speedup = old_time / new_time
```

4GHz, 1.5B instr: 50%ALU(1cy), 20%ctrl(3cy), 30%mem(2cy+40cy stall)
CPI_base = 0.5+0.6+0.6 = **1.7**
CPI_stall = 1.7+0.3×40 = **13.7** → 1.5B×13.7/4B = **5.14s**
stall→10: new CPI=1.7+3=4.7 → speedup=13.7/4.7=**2.91×**

1B instr, 25% mem, 40cy stall: base=1B, stall=0.25B×40=10B, total=11B
stall%=10/11=**90.9%** | stall→10: new=3.5B → speedup=11/3.5=**3.14×**

3GHz, 2B instr: 40%ALU(1), 30%mem(5), 30%ctrl(2): (no stall)
CPI=0.4×1+0.3×5+0.3×2=**2.5** | cycles=5B | time=5B/3GHz=**1.67s**
new: 20%ALU/50%mem/30%ctrl → CPI=0.2+2.5+0.6=**3.3** → time=**2.2s**

PIPELINING

- k stages, t = stage time. First: k×t. Each after: +t. Total n: (k+n−1)×t
- Purpose: ↑ throughput — NOT ↓ latency of one instruction
- Branch misprediction at end of stage j: flush stages 1..j−1 (wasted work)

5-stage, 10ms/stage, 10 instr: (5+9)×10 = **140ms**
Misprediction at end of stage 3 (instr 5): flushes 2 stages → +20ms penalty

AMAT — MEMORY HIERARCHY

2-Level (L1 + DRAM):

```
AMAT = L1_time + (1-L1_hit_rate) × miss_penalty
```

3-Level (L1 + L2 + DRAM):

```
AMAT = L1 + L1_miss × (L2 + L2_miss × DRAM)
P(reach DRAM) = L1_miss × L2_miss
NOTE: L2 hit rate is CONDITIONAL on L1 miss
```

L1=2ns(0.8hr), L2=12ns(0.75hr|L1miss), DRAM=100ns:
AMAT = 2 + 0.2×(12 + 0.25×100) = 2+0.2×37 = **9.4ns**
P(DRAM) = 0.2×0.25 = **5%**

L1=4ns(0.8), L2=20ns(0.75), DRAM=200ns:
AMAT = 4+0.2×(20+0.25×200) = 4+0.2×70 = **18ns**
L1→0.9: AMAT=4+0.1×70 = **11ns**

As avg cycles:

```
avg_cycles = hit_time + miss_rate ×
miss_penalty
total_time = num_accesses × avg_cycles /
clock_rate
```

2B accesses, hit=4cy, miss_pen=80cy, hr=0.85, 2GHz:
avg=4+0.15×80=**16cy** → time=2B×16/2B=**16s** | hr→0.95: **8s**, speedup=**2×**

UNIT CONVERSIONS — QUICK REF

```
Bandwidth:  1 Gbps  = 125 MB/s  (÷8 bits/byte)
            1.6 Gbps = 200 MB/s
            800 Mbps = 100 MB/s

Storage:    1 KB = 1,024 B    1 MB = 1,048,576
B
            1 GB = 10^9 B (problems use this)
OR  2^30 B
            → use 10^9 unless problem says 2^30

Time:       1 ms = 10^-3 s    1 µs = 10^-6 s
1 ns = 10^-9 s
            1 GHz clock = 1 cycle per ns

RPM→speed:  7200 RPM → period=60000/7200=8.33ms
→ avg_rot=4.17ms
            5400 RPM → period=11.11ms →
avg_rot=5.56ms

Bits:       1 byte=8 bits | addr space: n-bit
addr → 2^n bytes
```

CACHE LINES & LOCALITY

```
elem/line = cache_line_bytes / element_bytes
miss_rate (sequential) ≈ 1 / elem_per_line
```

64B line, float64(8B): 8 elem/line → miss≈1/8
64B line, float32(4B): 16 elem/line → miss≈1/16

Row-major (C/NumPy default):

- outer=rows, inner=cols → sequential → **fast** (spatial locality)
- outer=cols, inner=rows → stride=row_length → **cache thrashing**

DATASET SIZE & I/O

```
row_bytes = Σ col_sizes  [int64=8B, float64=8B,
int32=4B]
file_size = row_count × row_bytes
passes = epochs × num_learning_rates
total_IO = file_size × passes
```

500M rows, (UserID int64, Timestamp int64, Value float64): row=24B | file=500M×24=**12GB**
10 epochs × 3 LR = 30 passes → IO = 12×30=**360GB**

FEATURE EXPANSION (PAIRWISE INTERACTIONS)

```
new_cols = C(d,2) = d×(d-1)/2     [d = num
features, no squared terms]
expanded_cols = d + 1_label + C(d,2)

Step A: train on orig   → read orig × epochs ×
LR
Step B: expand features → read orig + write
expanded
Step C: train on expanded → read expanded ×
epochs × LR
Total IO = A + B + C
```

200M rows, d=4 features + 1 label, float64, 3LR × 4epochs:
C(4,2)=6 | orig: 5×8×200M=**8GB** | expanded: 11×8×200M=**17.6GB**
A=8×4×3=96GB | B=8+17.6=25.6GB | C=17.6×4×3=211.2GB
Total = **332.8 GB**

ROW-STORE USEFUL THROUGHPUT

```
useful_data = file_size × (needed_cols /
total_cols)
useful_throughput = useful_data / runtime
bandwidth_util = useful_throughput /
device_bandwidth
```

12GB, 120 equal cols, 1 col needed, 30s, SSD=600MB/s:
useful=12GB/120=100MB | throughput=100/30=**3.33MB/s** | util=3.33/600=**0.56%**

SCHEDULING METRICS

```
Turnaround = completion - arrival
Response   = first_cpu_time - arrival   ← NOT completion
Waiting    = turnaround - burst_time
```

Response ≠ turnaround. Response = when job FIRST gets CPU.

All 3 metrics worked: P1(burst=5,arr=0), P2(burst=3,arr=1), FCFS order
P1: 0→5. P2: 5→8.
P1: TAT=5-0=**5** | response=0-0=**0** | wait=5-5=**0**
P2: TAT=8-1=**7** | response=5-1=**4** | wait=7-3=**4**
Avg TAT=6 | Avg response=2 | Avg wait=2

ALGORITHM COMPARISON

- **FCFS:** simple, no starvation; worst avg TAT if long job first (convoy)
- **SJF:** optimal avg TAT (non-preemptive); starvation risk for long jobs
- **SRTF:** optimal avg TAT (preemptive); best for TAT, high overhead
- **RR:** best avg response time; fair; bad TAT if q too large or small

Response time ranking (best→worst): RR ≥ SRTF > SJF > FCFS
Turnaround ranking (best→worst): SRTF = SJF > RR > FCFS

SCHEDULING ALGORITHMS

FCFS: treat the ready queue as a literal FIFO — process at the front runs until done, then dequeue next. Convoy effect: one long job stalls all shorter ones behind it.
SJF (non-preemptive): when CPU becomes free, scan all READY jobs and pick the one with the shortest burst. Does NOT interrupt a currently running job even if something shorter arrives.
SRTF (preemptive SJF): on every new arrival (and on each completion), compare the newcomer's burst to the running job's REMAINING time. If newcomer is shorter → preempt immediately. Think: always running the minimum remaining time in the ready set.
Round Robin: cycle through jobs giving each a fixed quantum q. Unfinished job goes to back of queue. Good response time. If q too small → context-switch overhead dominates; if q too large → degenerates to FCFS.

CONTEXT SWITCH OVERHEAD

Each switch adds switch_cost before the next job starts. First job has no overhead.

FCFS P1(20),P2(5),P3(15),P4(10), all t=0, 1ms switch:
P1:0 | P2:21 | P3:27 | P4:43 → wait avg=91/4=**22.75ms**
SJF order P2,P4,P3,P1 → starts 0,6,17,33 → avg=**14ms**

FORK() OUTPUT

Child is a copy of parent at the fork point. Both continue from after fork().

W(A); fork(); W(B); if(child) wait(); W(C);
A: once (before fork) | B: twice (both parent+child)
C: twice (child first if parent waits, then parent)
→ A B B C C

SRTF SIMULATION — REVIEW

```
P1(burst=7,arr=0)  P2(burst=4,arr=2)
P3(burst=1,arr=4)

FCFS:  [P1:0-7][P2:7-11][P3:11-12]
       TAT: 7,9,8  avg=8

SJF:   [P1:0-7][P3:7-8][P2:8-12]  (P3 shorter at t=7)
       TAT: 7,4,10  avg=7

SRTF:  [P1:0-2][P2:2-4][P3:4-5][P2:5-7][P1:7-12]
  t=2: P2.burst(4) < P1.rem(5) → preempt
  t=4: P3.burst(1) < P2.rem(2) → preempt
  t=5: P2.rem(2) < P1.rem(5)  → P2 first
       TAT: P1=12,P2=5,P3=1  avg=6
```

ROUND ROBIN TRACE — REVIEW

```
P1=9, P2=5, P3=3, P4=7  all arr=0, quantum=3
[P1:0-3][P2:3-6][P3:6-9][P4:9-12]
[P1:12-15][P2:15-17][P4:17-20]
[P1:20-23][P4:23-24]
Done: P3=9, P2=17, P1=23, P4=24
Avg TAT = (9+17+23+24)/4 = 18.25
Avg response = (0+3+6+9)/4 = 4.5
  ← response = time of FIRST cpu slice, not completion
```

DISK I/O — HDD MECHANICS

Three components of every I/O:

- **Seek:** arm moves to correct track (1–20ms, avg ~4–8ms)
- **Rotational latency:** wait for sector to spin under head (avg = half rotation)
- **Transfer:** data moves off platter (~100s MB/s, small per-op)

```
rot_period  = 60,000ms / RPM
avg_rot     = rot_period / 2
T_IO        = seek + avg_rot + transfer
transfer    = data_size / bandwidth
bandwidth   = IOPS × IO_size
```

Sequential vs random: sequential can be 100–300× faster than random because contiguous blocks skip seeks and rotational waits.

DISK I/O — CONTIGUOUS VS CHUNKED

```
Contiguous file:
  T = 1×seek + 1×avg_rot + file_size/BW

Chunked access (n chunks of size s):
  chunk_transfer = s / BW
  T_per_chunk    = seek + avg_rot + chunk_transfer
  T_total        = n × T_per_chunk
```

Slide example — 10GB, 7200RPM, seek=8ms, BW=150MB/s, 512KB chunks:
rot=60000/7200=8.33ms → avg_rot=**4.17ms**
chunk_transfer=0.5/150=0.00333s=**3.33ms**
T_per_chunk=8+4.17+3.33=**15.5ms**
n_chunks=10240MB/0.5MB=**20,480**
Total=20480×15.5ms=317s≈**5.3min**
Sequential only: 10240/150=**68s** → chunked is **4.7× slower**

Contiguous vs fragmented — 2GB, 7200RPM, seek=8ms, BW=150MB/s, 512 frags:
avg_rot=4.17ms | transfer=2048/150=13.65s
Contiguous: 12.17ms+13.65s≈**13.66s**
Fragmented: 512×12.17ms+13.65s≈**19.88s** (**1.46× slower**)

STORAGE PERFORMANCE TERMS

- **IOPS:** I/O operations per second the device can sustain
- **Bandwidth:** bytes/sec = IOPS × IO_size
- **Latency:** total time to deliver data (seek+rot+transfer+OS delays)

40GB file, disk BW=800MB/s: $40\times10^9/8\times10^8$=**50s**
Same over 1.6Gbps network (=200MB/s): $40\times10^9/2\times10^8$=**200s**

SSD vs HDD: SSD has no seek/rotation → 100× faster random I/O, 10–100× higher BW. Still block-addressable.

DISK SCHEDULING

SSTF (Shortest Seek Time First): pick pending request closest to current head position. Fast but can starve far-away tracks.
SCAN/C-SCAN (elevator): sweep arm in one direction serving requests, then reverse (or jump back). Prevents starvation; ignores rotational latency.
SPTF (Shortest Positioning Time First): account for BOTH seek AND rotation — pick the request with least total positioning time. Best in practice; done inside drive firmware today.

MULTI-CORE SCHEDULING

SQMS: one shared queue, any core picks next. Simple; lock contention at scale.
MQMS: one queue per core. Less contention; risk of load imbalance.

```
Makespan = time until last job finishes = max(sum of jobs per core)
```

ROW-STORE VS COLUMN-STORE

Row-store: all columns of a row stored together. Good for full-row reads (OLTP). Bad for column-only queries — must scan all columns.
Column-store (Parquet, ORC): each column stored together. Great for column-subset queries — only read needed columns. Better compression (same-type data together).
Parquet vs CSV: compressed (up to 10× smaller), column pruning, schema+metadata embedded, immutable. Cannot edit in-place.

1TB CSV → 130GB Parquet; query 236s→6.78s (34×); scans 99% less data.

PROCESS STATES (QUICK REF)

NEW → READY → RUNNING → BLOCKED → READY

- RUNNING→BLOCKED: process issues I/O syscall
- **BLOCKED→READY: I/O completes**
- RUNNING→READY: timer interrupt (preemption)
- Threads share address space; processes don't
- GIL: CPython = 1 thread at a time (CPU-bound bottleneck)

OS MECHANICS — HOW IT ACTUALLY WORKS

User mode vs Kernel mode: Hardware enforces two privilege levels. Normal programs run in user mode (can't touch hardware directly). OS runs in kernel mode (full access).
Syscall (system call): only way for user program to request OS services. Program executes a **trap** instruction → CPU switches to kernel mode → OS handles the request → returns to user mode. Examples: read(), write(), fork(), malloc() (which calls brk/mmap).
Timer interrupt: hardware timer fires ~every 10ms. CPU automatically jumps to OS interrupt handler (in kernel mode). OS can then: (1) preempt current process (RUNNING→READY), (2) pick another process, (3) return. This is how the OS regains control without process cooperation.
Context switch — step by step:

1. Timer fires → CPU saves PC + registers onto kernel stack
2. OS saves them into the outgoing process's **PCB**
3. Scheduler picks next process from ready queue
4. OS loads new process's PCB (registers, PC) → switch page table (new VA space)
5. Return from interrupt → new process resumes in user mode

PCB (Process Control Block): per-process OS struct storing registers, PC, state, open files, memory map, PID. Lives in kernel memory.

VIRTUAL MEMORY BASICS

```
offset_bits = log₂(page_size)
VPN_bits    = address_bits − offset_bits
# pages     = 2^(VPN_bits)  =  VA_space /
page_size

VPN    = virtual_addr ÷ page_size  (upper bits)
offset = virtual_addr mod page_size (lower
bits)

physical_addr = PFN × page_size + offset
```

8-bit VA, page_size=16B (2^4):
offset_bits=4, VPN_bits=4, pages=2^4=16
VA=35: VPN=35÷16=**2**, offset=35 mod 16=**3**
If PFN for VPN 2 = 7 → PA = 7×16+3 = **115**
VA=50: VPN=3, offset=2, PFN=1 → PA=1×16+2 = **18**

32-bit VA, 4KB page: offset=12 bits, VPN=20 bits
entries per process = 2^20 = **1,048,576**

PAGE TABLE SIZE

```
Step 1: offset_bits = log₂(page_size)
Step 2: VPN_bits = address_bits − offset_bits
Step 3: entries per process = 2^(VPN_bits)
Step 4: one_PT_bytes = entries × PTE_size
Step 5: all_PT_bytes = one_PT_bytes ×
num_processes
Step 6: frames_needed = ceil(all_PT_bytes /
page_size)
```

Why frames? Page tables themselves live in physical memory. Each frame holds page_size bytes of page table data.

Worked (the "4 frames" example):
20-bit VA, page=4KB, PTE=4B, 16 processes
Step 1: offset = $\log_2(4096)$ = **12 bits**
Step 2: VPN = 20 − 12 = **8 bits**
Step 3: entries/process = 2^8 = **256**
Step 4: 1 PT = 256 × 4B = **1024 bytes**
Step 5: all PTs = 1024 × 16 = **16,384 bytes**
Step 6: frames = 16384 / 4096 = **4 frames**

24-bit VA, 4KB page, PTE=4B, 8 processes:
VPN=12 bits, entries=4096, 1PT=16KB, all=128KB → frames=128KB/4KB=**32**

VA_space=2^16B, page=2^8B:
offset=8, VPN=8, entries=256/process

PAGE REPLACEMENT POLICIES

FIFO: evict page loaded EARLIEST (track load order)
LRU: evict LEAST RECENTLY USED — update on every HIT too, not just faults
OPT: evict page whose NEXT USE is farthest away (theoretical best)
Simulation steps:

1. Go through reference string left→right
2. HIT: page in frame → no fault; update LRU order
3. FAULT: page not in frame → if empty slot, fill it; else evict per policy
4. Count total faults and hits

PAGE REPLACEMENT EXAMPLE

Reference string: 1 2 3 4 1 2 5 1 2 3 4 5 — 3 frames
FIFO (evict oldest loaded):

```
Ref: 1  2  3  4  1  2  5  1  2  3  4  5
     F  F  F  F  F  F  F  -  -  F  F  -
Frames after each (queue order):
1→[1]  2→[1,2]  3→[1,2,3]  4→evict1[2,3,4]
1→evict2[3,4,1]  2→evict3[4,1,2]
5→evict4[1,2,5]
1=hit  2=hit  3→evict1[2,5,3]  4→evict2[5,3,4]
5=hit
Faults = 9
```

LRU (evict least recently used):

```
Ref: 1  2  3  4  1  2  5  1  2  3  4  5
     F  F  F  F  F  F  F  H  H  F  F  F
After 4: evict LRU=1 → [2,3,4]
After 1: 1∉{2,3,4} → evict LRU=2 → [3,4,1]
After 2: 2∉{3,4,1} → evict LRU=3 → [4,1,2]
After 5: evict LRU=4 → [1,2,5]
1,2: hits | 3: evict LRU=5 → [1,2,3] | 4: evict
LRU=1 → [2,3,4] | 5: F
Faults = 10
```

OPT (evict page used farthest in future):

```
Ref: 1  2  3  4  1  2  5  1  2  3  4  5
     F  F  F  F  H  H  F  H  H  F  F  H
At 4: [1,2,3]. Next: 1@+1, 2@+2, 3@+6 → evict 3
→ [1,2,4]
At 5,6: HIT (1,2 in frames)
At 7: [1,2,4]. Next: 1@+1, 2@+2, 4@+4 → evict 4
→ [1,2,5]
At 8,9: HIT (1,2)
At 10: [1,2,5]. 1,2→never again, 5@+2 → evict 1
→ [2,5,3]
At 11: [2,5,3]. 2,3→never, 5@+1 → evict 2 →
[5,3,4]
At 12: ref=5 → HIT
Faults = 7
```

Key: Thrashing = more page faults than useful work → CPU util drops. Fix: reduce multiprogramming, add RAM.

Larger pages: ↑ internal fragmentation (waste within page). Paging = no external fragmentation.

ABCADBEABC EXAMPLE (3 FRAMES)

Reference string: A B C A D B E A B C
FIFO:

```
A→[A]F  B→[A,B]F  C→[A,B,C]F  A=hit
D→evict A[B,C,D]F  B=hit  E→evict B[C,D,E]F
A→evict C[D,E,A]F  B→evict D[E,A,B]F  C→evict
E[A,B,C]F
Faults = 8
```

LRU:

```
A→[A]F  B→[A,B]F  C→[A,B,C]F  A=hit→order:
[B,C,A]
D→evict B(LRU)[C,A,D]F  B→evict C[A,D,B]F
E→evict A[D,B,E]F
A→evict D[B,E,A]F  B=hit  C→evict E[A,B,C]F
Faults = 8  (tie with FIFO here)
```

OPT:

```
A→[A]F  B→[A,B]F  C→[A,B,C]F  A=hit
D: next use: B@+1,A@+3,C→never → evict C →
[A,B,D]F
B=hit  E: D→never → evict D → [A,B,E]F
A=hit  B=hit  C: all→never → evict any →
[A,B,C]F
Faults = 6  (OPT beats FIFO=LRU=8 on this
string)
```

VPN/PFN TRANSLATION (MORE EXAMPLES)

16-bit VA, page=256B (2^8): offset=8 bits, VPN=8 bits
VA=0x12AB: VPN=0x12=18, offset=0xAB=171
If page table[18]=PFN 5: PA=5×256+171=**1451**
Page size = 4KB = 2^12: offset = 12 bits always for 4KB
VA_bits=32: VPN=20 bits, entries=2^20 per process

PTE size: must hold PFN + control bits (valid, dirty, read/write)
If PA space = 2^20 bytes, 4KB pages → 2^8 physical frames → need 8 bits for PFN + overhead → round up to 4B per PTE

PAGE OFFSET / VPN BITS PRACTICE

```
Given: VA_space size S, page size P
  → offset = log₂(P)
  → VPN = log₂(S/P) = log₂(S) − log₂(P)

Given: 32KB VA, 512B pages:
  → offset = log₂(512) = 9 bits
  → VPN = log₂(32768/512) = log₂(64) = 6 bits
  → 64 virtual pages
```

VM TRANSLATION FLOW (CONCEPTUAL)

Why VM exists: Each process gets illusion of its own private address space (isolation). OS+hardware prevent one process from touching another's memory. Allows using disk as overflow when RAM is full.

Every memory access — what happens:

1. CPU generates virtual address (VA = VPN + offset)
2. MMU checks **TLB** (hardware cache of recent VPN→PFN translations, ~1 cycle)
3. **TLB HIT:** get PFN → PA = PFN × page_size + offset. Done.
4. **TLB MISS:** MMU walks page table in memory → load PFN into TLB → retry
5. **Page FAULT:** PTE says "not in RAM" → trap to OS → OS finds page on disk → loads into free frame → updates PTE → TLB loaded → resume

TLB: tiny fast associative cache on the CPU die (~32–1024 entries). Stores recently used VPN→PFN pairs. Flushed on context switch (different process = different VA space).

Page fault cost: disk access = millions of cycles. So thrashing (constant faults) kills performance.

Valid bit in PTE: 0 = page not in RAM (trigger fault). 1 = page present, PFN valid. **Dirty bit:** 1 = page was written → must write back to disk before evicting.

AMDAHL'S LAW

```
speedup(n) = 1 / ((1-p) + p/n)
  p = parallelizable fraction
  n = number of workers/cores

max_speedup (n→∞) = 1 / (1-p)

speedup = T_serial / T_parallel
efficiency = speedup / n_workers
```

p=0.75 (75% parallel), n=8 cores:
max = 1/(1−0.75) = 1/0.25 = **4×**
8-core = 1/(0.25 + 0.75/8) = 1/0.34375 = **2.91×**
5-core = 1/(0.25 + 0.75/5) = 1/0.4 = **2.5×** | single-core 80min → **32min**

p=0.9, n=10: speedup=1/(0.1+0.09)=1/0.19=**5.26×**
max=1/0.1=**10×** | efficiency=5.26/10=**52.6%**

T_serial=24min, T_parallel(6 workers)=5min:
speedup=24/5=**4.8×** | efficiency=4.8/6=**80%**

Serial part=10s, parallel=90s, n=9: new_parallel=10s
T_new=10+10=20s | speedup=100/20=**5×**

TASK GRAPH LOWER BOUNDS

```
work       = sum of all task durations
crit_path  = longest path from start to end
(serial deps)
lower_bound = max(work / n_workers, crit_path)
max_speedup = work / lower_bound
idle_time   = n_workers × actual_makespan -
work
```

work=100, crit_path=40, 4 workers:
LB = max(100/4, 40) = max(25, 40) = **40**
max_speedup = 100/40 = **2.5×**
actual makespan=50: idle=4×50−100=**100 worker-units**

work=60, crit=15, 3 workers:
LB=max(60/3,15)=max(20,15)=**20**
max_speedup=60/20=**3×** (linear! possible here)

Spring 26-2: work=70, crit=35 (T1→T4→T6), T3=10 alone
3W: LB=max(70/3,35)=35 → speedup=**2×**, idle=3×35-70=**35**
2W: LB=max(70/2,35)=35 but actual makespan=**40**
(T6 blocked until BOTH T4 and T5 finish → forces extra 5 units)

Critical path = serial bottleneck. Speedup ≤ work/crit even with ∞ workers.

CPU UTILIZATION / EFFICIENCY

```
CPU_util = compute_time / (compute + wait_time)
efficiency = useful_work / total_work_done
```

18s compute, 27s communication overhead:
util = 18/(18+27) = 18/45 = **40%**

Job: 5s useful, 15s total (incl. sync + idle):
efficiency = 5/15 = **33%**

NUMA & FALSE SHARING

NUMA (Non-Uniform Memory Access):

- Multi-socket machines: each socket has "local" RAM
- Accessing remote socket's RAM = k× slower than local

Local=100ns, remote=3× slower:
Thread on Socket 0 reading Socket 1's memory → **300ns**

False Sharing:

- Two threads write **different variables** on the **same cache line**
- Every write forces other core to invalidate its copy → constant line bouncing
- Fix: pad struct so each thread's variable is on its own cache line

Thread A writes x, Thread B writes y. x and y are adjacent ints (4B each) on same 64B cache line → false sharing → repeated cache invalidations even though they write to different variables.

True sharing: threads actually share the same variable (needed for coordination).

PARALLELISM CONCEPTS

SIMD: single instruction, multiple data — one instruction on a vector of values (e.g., add 8 floats at once). Used in NumPy, GPU.

GPU vs CPU: GPU = thousands of simple cores for data-parallel work; CPU = few powerful cores for complex branchy code.

Why NumPy beats Python loops: contiguous typed memory, compiled C kernels, SIMD vectorization, no per-element Python overhead.

Parallel overhead types: (1) **Communication:** cores share results (network/memory bus). (2) **Synchronization:** waiting at barriers/locks. (3) **Load imbalance:** fast workers idle waiting for slow ones. All reduce efficiency below 1.0.

LOCKS & SYNCHRONIZATION

Race condition: two threads read/modify same shared data concurrently. Outcome depends on timing. Classic: both read x=0, both do x++, both write x=1 → lost update (should be 2).

Critical section: code region that must run atomically — only one thread at a time. Solution: protect with a lock.

Mutex (lock): binary flag. thread calls **lock()** to enter — if already held, the caller **sleeps** (blocks). Holder calls **unlock()** when done → one waiter wakes up.

Spinlock: thread busy-loops checking the lock (burns CPU). Use when wait is very short (saves context-switch overhead). Bad if wait is long.

Deadlock: Thread A holds Lock 1, needs Lock 2. Thread B holds Lock 2, needs Lock 1. Both block forever.

Fix deadlock: always acquire locks in the **same global order** (e.g., always lock 1 before lock 2).

Atomic operations: hardware guarantee — read-modify-write completes without interruption (compare-and-swap, fetch-and-add). No lock needed for simple counters.

Condition variable: lets a thread sleep until a condition is true (e.g., buffer non-empty). Used with a mutex: unlock mutex → sleep → re-lock on wakeup.

TASK GRAPH EXAMPLE

Tasks with dependencies — draw the DAG:

```
A=3, B=2, C=4, D=1, E=5
Deps: B after A, C after A, D after B+C, E
after D

Paths:
A→B→D→E = 3+2+1+5 = 11
A→C→D→E = 3+4+1+5 = 13  ← critical path

work = 3+2+4+1+5 = 15
crit_path = 13
2 workers: LB = max(15/2, 13) = max(7.5, 13) =
13
max_speedup = 15/13 ≈ 1.15×

Schedule on 2 workers:
W1: A(0-3) → C(3-7) → D(7-8) → E(8-13)
W2: idle(0-3) → B(3-5) → idle(5-8)
Makespan = 13, idle = 2×13-15 = 11 worker-units
```

AMDAHL: SOLVING FOR P OR N

```
Given speedup S and n workers: solve for p
  S = 1/((1-p) + p/n)
  1/S = (1-p) + p/n = 1 - p + p/n = 1 - p(1 -
1/n)
  p = (1 - 1/S) / (1 - 1/n)

Given speedup S and p: solve for n
  1/S = 1 - p + p/n
  p/n = 1/S - 1 + p
  n = p / (1/S - 1 + p)
```

Observed speedup=3× with 6 cores: p=(1−1/3)/(1−1/6)=(2/3)/(5/6)=**0.8** (80% parallel)

STORAGE / THROUGHPUT SUMMARY

```
Disk: random I/O = seek + rot + transfer  (very
slow)
SSD:  no seek/rot — ~10-100× faster random I/O
than disk
DRAM: ~10-100× faster than SSD
Cache: ~10× faster than DRAM

Useful BW = data_needed /
time_to_read_everything
  (row-store reading 1 of N cols → useful BW =
device_BW / N)
```

MCQ CONCEPT KEYS

- **Scheduling goal:** maximize CPU util / multiprogramming — NOT speed up one process
- **Response time low =** important for **interactive** jobs. Batch jobs → minimize turnaround
- **Processes don't share** virtual addr space. DO share: physical RAM, CPU, disk
- **Cache hit rate** primarily set by cache **SIZE** (larger → fewer misses). Not DRAM speed
- **Timer interrupt** = how OS reclaims CPU from running process without its cooperation
- **"2's complement of 12"** = representation of −12: flip(00001100)+1 = **11110100**
- **ABCD∞ LRU 3 frames:** 100% fault rate — 4 unique pages > 3 frames, always evicts
- **ABCD∞ OPT 3 frames:** ≈33% fault rate — OPT can load ABC, strategically defer D
- **fork() child:** child=0 in child process, child=PID in parent. Both resume after fork()
- **Larger pages:** ↑ internal frag, ZERO external frag (paging always uses fixed-size frames)
- **Thrashing** → ↓ CPU utilization (CPU spending all time on page faults, not useful work)
- **SJF:** selects shortest at time CPU is free. Does NOT preempt running job even if shorter arrives
- **SRTF preempts** when new arrival's burst < current job's REMAINING time (not original burst)